Membrane Transport: Moving Across the Bilayer

Week 8 · Reading: Chapter 7.2–7.5

- Why the plasma membrane is **selectively permeable**
- Passive transport: diffusion, facilitated diffusion, osmosis
- Active transport: pumps and cotransport
- Bulk transport: endocytosis and exocytosis

Quiz 5 (Tue 10 AM) covers this lecture and Lab 4.

The Fluid Mosaic Model

- Phospholipid bilayer: hydrophilic heads face water, hydrophobic tails face inward
- **Integral proteins** span the membrane (channels, carriers, pumps); **peripheral proteins** sit on the surface
- **Cholesterol** = fluidity buffer: keeps the membrane fluid when cold, stable when warm
- **Glycoproteins / glycolipids** = cell recognition (e.g., blood types A, B, O)

Exam tip: small nonpolar molecules (O_2, CO_2) cross the bilayer directly; ions and polar molecules need proteins.

Tonicity: Which Way Does Water Move?

Hypotonic solution
→ animal cell swells, may **lyse**
plant cell: **turgid** (ideal)

Isotonic solution
→ no net movement
animal cell: normal

Hypertonic solution
→ animal cell **crenates**
plant cell: **plasmolysis**

Water moves toward the **higher solute** concentration. In Lab 4, the potato cores lose mass in sucrose solutions above their isotonic point (≈0.3 M in most sections).

Active Transport

Mechanism	Energy source	Example
Sodium-potassium pump	ATP directly (1 ATP per cycle)	3 Na^+ out, 2 K^+ in; builds membrane potential
Proton pump	ATP	Plants: H^+ gradient powers other transport
Cotransport (secondary active)	An existing ion gradient	H^+/sucrose cotransporter loads sugar into phloem

Common mistake: cotransporters do not use ATP directly; the pump that built the gradient did.

Bulk Transport & Review

- **Exocytosis:** vesicles fuse with the membrane to release contents (insulin, neurotransmitters)
- **Phagocytosis:** "cell eating" (white blood cells engulf bacteria)
- **Pinocytosis:** "cell drinking", nonspecific uptake of fluid
- **Receptor-mediated endocytosis:** specific uptake (LDL cholesterol); defects cause familial hypercholesterolemia

Practice: (1) Why does a cell need ATP to move Na^+ out? (2) Predict what happens to a red blood cell in pure water. (3) Name the transport type for O_2 entering a cell.